View Diff: 3D-Consistent Image Generation with Test-to-Image Models

In Meta Reality Labs

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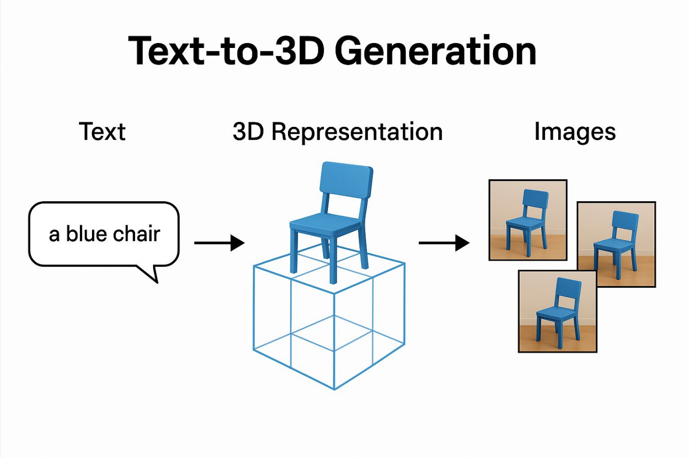
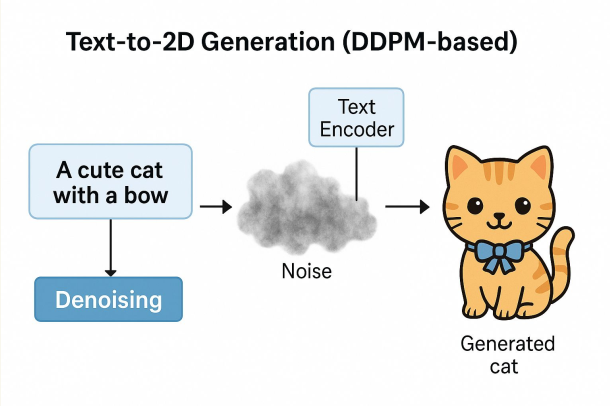
Lorem Ipsum



- Existing text–to–3D methods use text–to–image diffusion models in an optimization problem or fine–tune them on synthetic data.
- In this paper, a method is presented, that leverages pretrained text-to-image models as a prior and learns to generate multi-view images in a single denoising process from real-world data
- Proposed to integrate 3D volume-rendering and cross-frame-attention layers into each block of the existing U-Net network of the text-to-image model.
- an autoregressive generation that renders more 3D-consistent images at any viewpoint

Related Work:



Text-to-2D Generation (DDPM-based):

- Uses denoising diffusion probabilistic models (DDPM) that learn to reverse a Gaussian noise process. It has proven to generate high-quality images and outperforms older methods like GANs.
- Large-scale text-conditioned models (e.g., Imagen, DALL-E 2) are built on these principles.
- Conditional generation can be achieved via classifier guidance or classifier-free guidance.
- Techniques like ControlNet allow further conditioning (e.g., on segmentation maps) to tune the output.
- Essentially, a pretrained text-to-image (T2I) model converts text prompts into detailed 2D images.

Text-to-3D Generation:

- Extends 2D DDPMs to create 3D shapes or scenes from text.
- Methods like DreamFusion use score distillation sampling (SDS) to optimize a 3D shape whose renders align with the 2D DDPM's outputs.
- Second-stage mesh optimization can improve quality, though sometimes at the cost of view consistency.
- Some approaches condition on an input image plus camera motion to generate novel views, but these may lack explicit geometry, leading to inconsistent views.
- This approach fine-tunes a 2D T2I model and adds explicit 3D unprojection and rendering operators to enforce 3D consistency directly.

Diffusion on 3D Representations:

- Some methods directly model 3D distributions using available 3D data (e.g., DiffRF uses ground-truth 3D shapes).
- Others, like HoloDiffusion, are supervised only with 2D images and then render a 3D shape.
- The challenge is that 3D datasets are much smaller compared to 2D ones, limiting diversity.
- By leveraging a large 2D pretrained DDPM and adding 3D-specific components tuned on multi-view data, the model achieves better multi-view consistency while keeping the expressive power of large-scale image training.

Method:

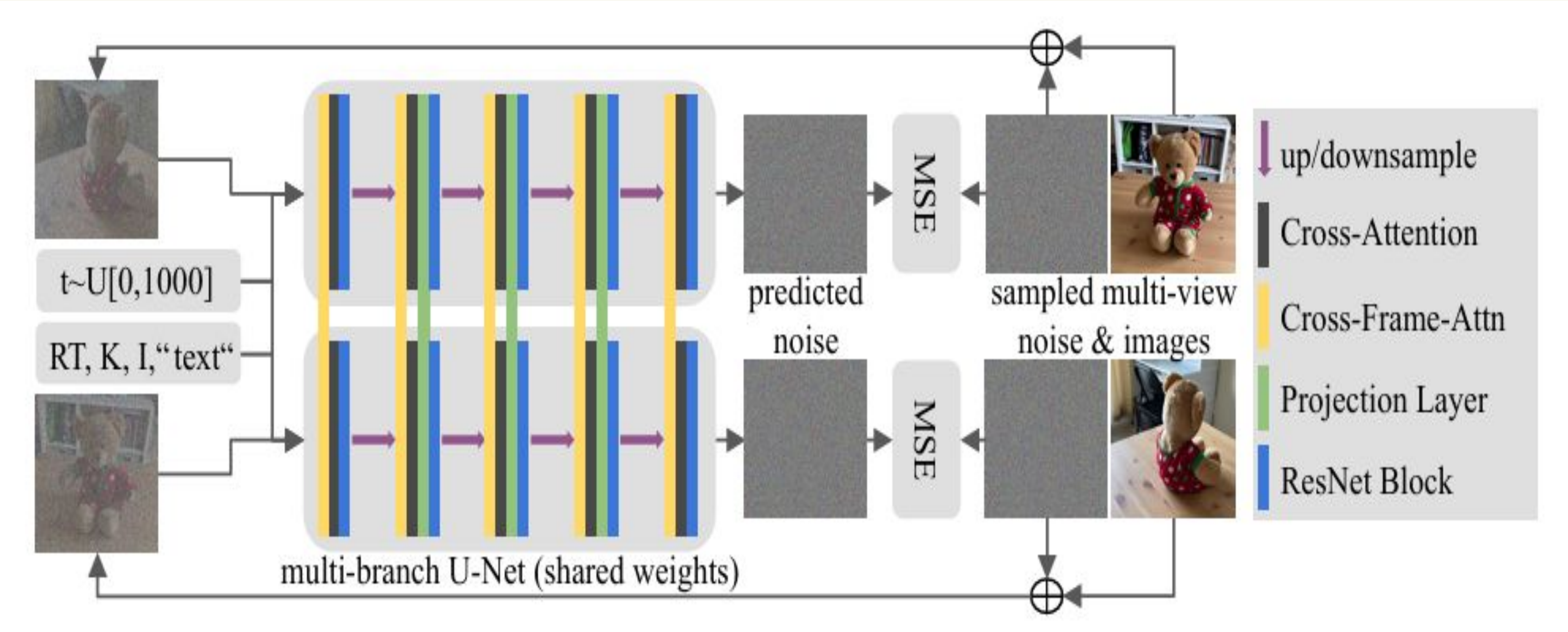
- ➢ Method takes a text prompt or an input image along with desired camera poses, and then generates a set of images that all represent the same object consistently from different viewpoints. Here's a simple breakdown.
- ➢ This approach leverages a strong T2I model, fine-tunes it with multi-view data, and adds 3D-aware layers to generate a set of images that are consistent in 3D—allowing you to view the same object from any angle without losing detail or coherence.


a stuffed bear sitting on a wooden box

Method:

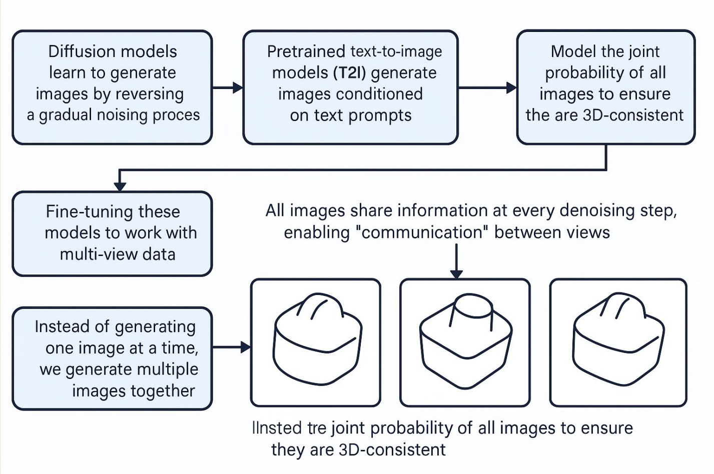
- Start with Gaussian noise sampled separately per image, then denoise them through Neural Network $\mu\theta$, there is communication between images to use previous state
- Each image's denoised output x_n^{t-1} depends on the entire set of current noisy images $x_{0:N}^t$.
- The model **shares a neural network μ_θ** across all views to predict the mean of the denoised image.
- To model a 3D-consistent denoising process over all images, we predict per-image noise through a neural network $\epsilon\theta$. This neural network is initialized from the pretrained weights of existing text-to-image models

Method:



t~U[0,1000]
RT, K, I,“text“
multi-branch U-Net (shared weights)
predicted noise
MSE
MSE
sampled multi-view noise & images
up/downsample
Cross-Attention
Cross-Frame-Attn
Projection Layer
ResNet Block

3D–Consistent Diffusion:



- Diffusion models learn to generate images by reversing a gradual noising process.
- They model the probability distribution of data by predicting the noise added to an image
- Pretrained text-to-image models (T2I) generate images conditioned on text prompts.
- In this method**, fine-tuning these models to work with multi-view data.**
- Instead of generating one image at a time, we generate multiple images together.
- Model the joint probability of all images to ensure they are 3D-consistent.
- All images share information at every denoising step, enabling “communication” between views.

- The generation process **starts from Gaussian noise** for each image.
- At each step, a neural network predicts the next denoised state, conditioned on the current states of all images.
- This step-by-step process makes sure that each new image remains consistent with the others.
- The model learns to predict the noise added during the forward (noising) process.
- An L2 loss is used to train the noise predictor, ensuring accurate and consistent denoising across all views.
- Added two Layers to existing U-Net

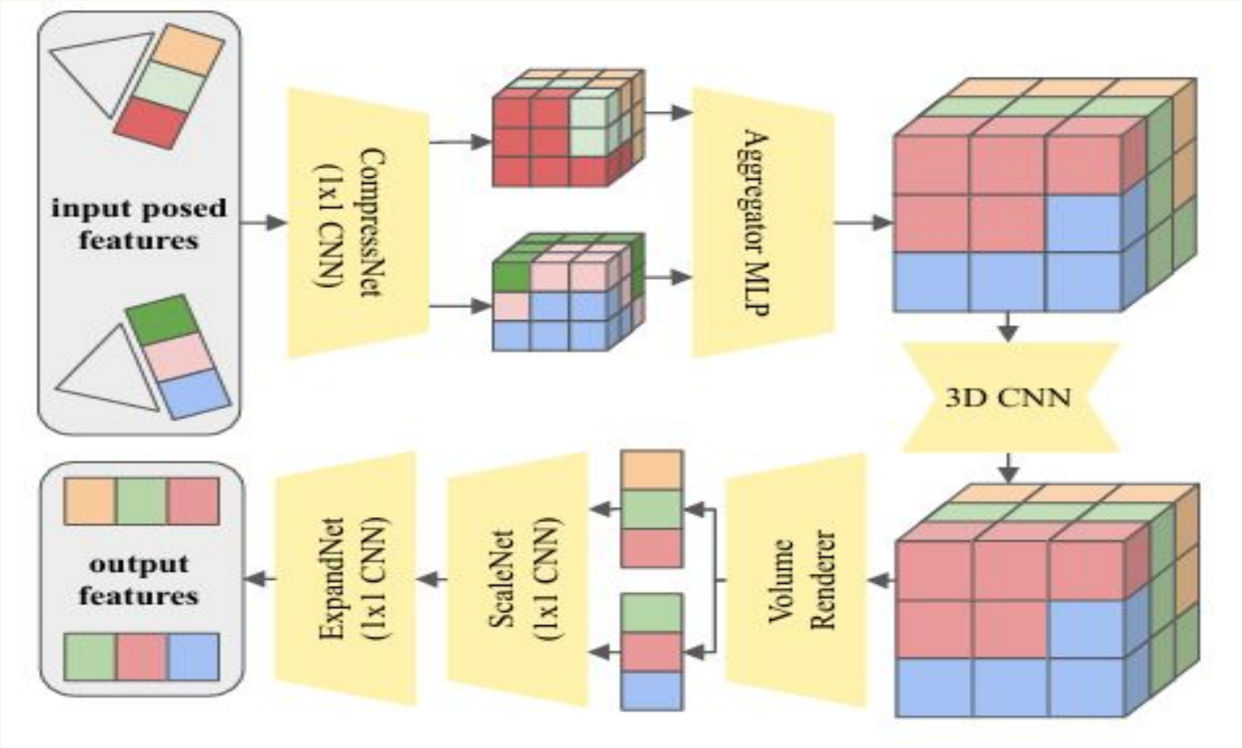
Cross-Frame Attention

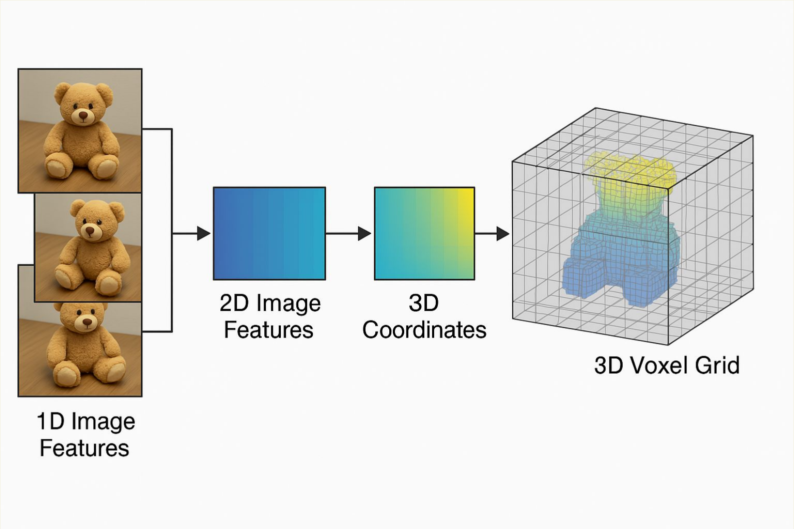
- Instead of treating each image separately, cross-frame attention allows the network to compare and align features across multiple images (or frames).
- It modifies the standard self-attention layer to compute attention across all images, using formulas like:
  - $CFAttn(Q, K, V) = softmax(QK^T/\sqrt{d}) \cdot V$
- Here, Q, K, V are derived from each image's spatial features using pretrained weights.
- A conditioning vector is added to inform the network about each image's viewpoint.
  - This vector includes:
    - **Pose information** (from the camera matrix)
    - **Camera details** (focal length and principal point)
    - **Intensity encoding** (mean and variance of RGB values)
- The conditioning is incorporated via extra linear layers, ensuring that images from different views maintain a consistent global style.

Projection Layer

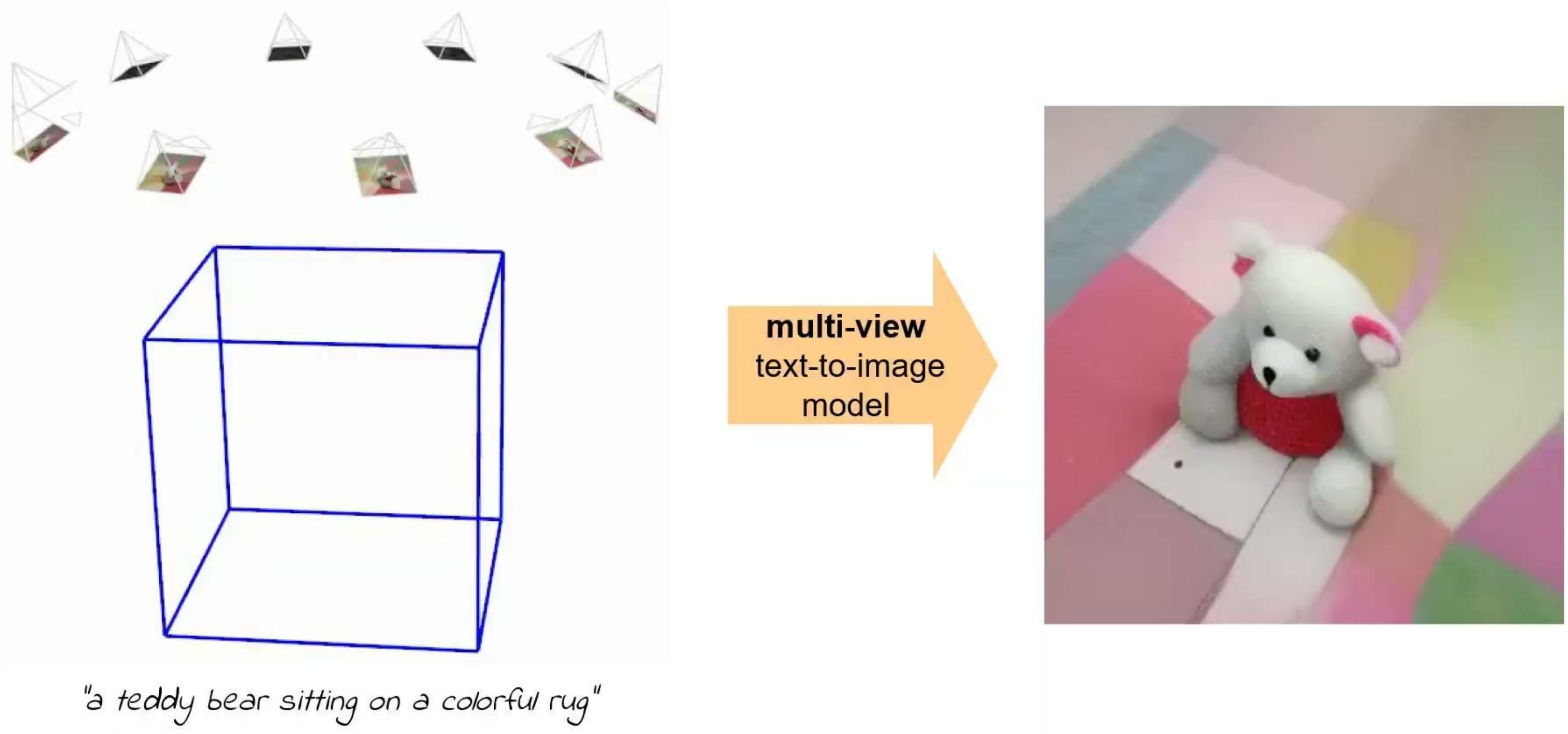
- Ensures that the generated images align in 3D space and follow the specified camera poses.
- Cross-frame attention helps match features globally, but objects **might not strictly adhere** to the desired poses, leading to inconsistencies.
- A projection layer is added into the U-Net architecture (except at the very first and last blocks).
- It converts 2D image features into a unified 3D voxel grid, creating 3D-consistent features.
- These features are then refined by subsequent layers (e.g., ResNet blocks) to enforce 3D coherence.
- Draws on techniques from multi-view stereo to aggregate spatial features across views.

Projection Layer





- Converts compressed 2D image features into 3D coordinates.
- Use an MLP to combine features from multiple views into a single voxel grid.
- Process the voxel grid with a 3D CNN to enhance and refine the features.
- Render the refined voxel grid into 3D-consistent features using a NeRF-like volume renderer; apply half of the background to MERF
- Apply a learned scaling function and expand the feature dimensions to match the desired output.


multi-view
text-to-image
model
"a teddy bear sitting on a colorful rug"

Autoregressive Generation

- The model takes several images (samples) at once (e.g., N=5 during training, up to N=30 at inference).
- All samples are denoised together to ensure they remain 3D-consistent across different views.
- Feature viewpoints are generated based on previously produced images.
- The network uses the denoising timestep inputs for all images, then varies these timesteps to condition the next set of outputs.

Unconditional Generation:

- Start with all samples as Gaussian noise.
- Use the same timestep for each sample.
- Provide different camera parameters and a single text prompt.
- The result is a set of images showing the object from various viewpoints in a consistent 3D manner.

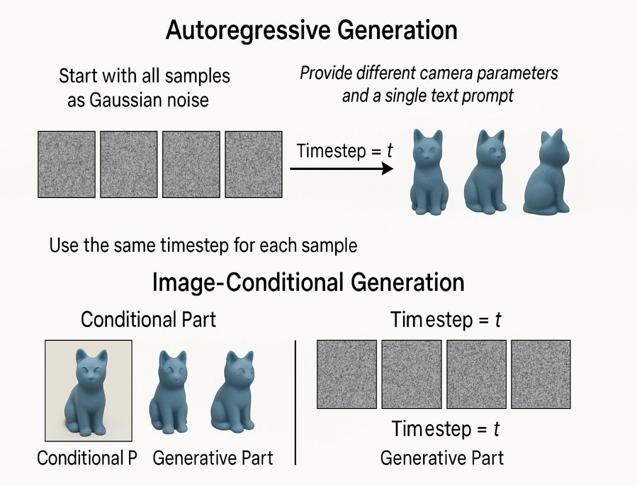


Image-Conditional Generation:

- Split samples into two groups:
  - **Conditional Part (nc):** Given images and cameras (e.g., one input image).
  - **Generative Part (ng):** Initialized from noise, then conditioned on the provided images.
- For the conditional samples, set their timestep to zero (i.e., no noise) to guide the generation.This setup allows the model to reconstruct from a single image or generate novel views smoothly.

Comparing results

ViewsetDiffusion and HoloFusion are other fusion models used for 3D generation

Dataset: co3d data set by meta


sample 1
sample 2
[C] wearing a green hat
[C] sits on a green blanket
sample 1
sample 2
rusty [C] in the grass
yellow and white [C]

Limitations:

- ○ Some generated images show slight inconsistencies (e.g., exposure changes) across viewpoints due to view-dependent effects in the scene
- ○ Current work primarily focuses on individual objects, not entire scenes.

Potential Improvements:

- **Lighting Control:**
  - Incorporate lighting condition control (e.g., via ControlNet) to reduce view-dependent inconsistencies.
- **Scene-Scale Generation:**
  - Explore extending the method to generate full scenes using larger datasets.

Thank You